AI Fundamentals and Workflow

Definitions, core concepts, AI pipeline, and RAG architecture

1. What is Artificial Intelligence (AI)?

Artificial Intelligence is the ability of computer systems to perform tasks that usually require human intelligence, such as understanding language, recognizing patterns, learning from data, and solving problems.

2. Key Concepts

Data: Information used to train or run AI models.
Algorithm: Rules or methods used to identify patterns and make predictions.
Model: A program that learns patterns and produces predictions or decisions.
Training: The process of teaching a model using data.
Inference: Using a trained model on new inputs.
Evaluation: Measuring model quality using suitable metrics.

3. Types of AI

Narrow AI: Designed for specific tasks, such as chatbots or recommendations.
General AI: A hypothetical AI with broad human-like capabilities.
Generative AI: Creates new text, images, audio, video, or code.

4. Important Terms

ML: Learns patterns from data without explicit rules for every case.
Deep Learning: ML using multi-layer neural networks.
NLP: Helps computers process human language.
LLM: A large language model trained to understand and generate text.
Embedding: A numerical representation of content used to compare meaning.
Hallucination: A plausible-sounding but incorrect or unsupported AI output.

5. AI Workflow

Data collection → **Data preprocessing** → **Model training** → **Evaluation** → **Deployment** → **Monitoring.** Collect relevant data, clean and transform it, train a model, test performance, make it available through an application or API, and monitor its quality over time.

6. RAG (Retrieval-Augmented Generation)

RAG combines an LLM with information retrieved from external documents or a knowledge base. It can help produce answers that are more relevant to the supplied context.

Typical RAG pipeline: Load documents → split into chunks → create embeddings → store in a vector database → retrieve relevant chunks for a user query → add context to the prompt → generate a response.

7. Real-World Applications

Chatbots and virtual assistants • Recommendation systems • Image and speech recognition • Healthcare support • Fraud detection • Document search • Education • Content generation.

Graph notes

The next page includes a visual infographic with AI workflow diagrams and illustrative graphs. Graph values are examples for learning, not measured results.

Key idea: AI workflow describes how a system is built and operated; RAG adds retrieval from a knowledge source at question time.

AI Fundamentals and Workflow

A Complete Guide

1 What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) refers to the ability of machines or computer systems to perform tasks that typically require human intelligence, such as understanding language, recognizing patterns, learning from data, making decisions, and solving problems.

2 Key Concepts in AI

- **Data:** Raw facts and information used to train or run AI models.
- **Algorithm:** Set of rules or methods to find patterns and make predictions.
- **Model:** A program that learns from data and makes predictions or decisions.
- **Training:** Process of teaching a model using data.
- **Inference:** Using a trained model to make predictions on new data.
- **Evaluation:** Measuring how well the model performs (accuracy, precision, recall, etc.).

3 Types of AI

- **Narrow AI (Weak AI):** Designed for specific tasks (e.g., chatbots, recommendation systems).
- **General AI (Strong AI):** Hypothetical AI with human-level intelligence (still under research).
- **Generative AI:** Creates new content such as text, images, code, audio, etc. (e.g., ChatGPT, DALL·E).

4 Important AI Terms

- **Machine Learning (ML):** Learns from data without explicit programming.
- **Deep Learning (DL):** Subset of ML using neural networks to learn complex patterns.
- **Natural Language Processing (NLP):** Helps machines understand and process human language.
- **Computer Vision (CV):** Helps machines understand images and videos.
- **Large Language Model (LLM):** A type of deep learning model trained on large text data.
- **Generative AI:** AI that generates new content (text, image, audio, code, etc.).
- **RAG (Retrieval Augmented Generation):** Combines external knowledge with LLM for accurate and up-to-date responses.

5 AI Workflow (Overview)

Data Collection → Data Preprocessing → Model Training → Evaluation → Deployment → Real-World Use

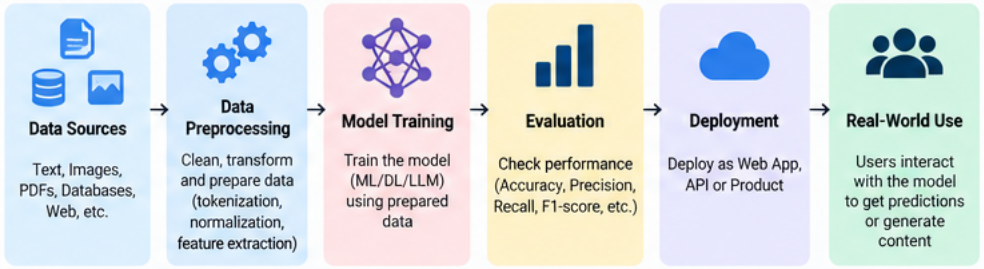
6 Real-World Applications

- Chatbots and Virtual Assistants
- Recommendation Systems
- Image and Speech Recognition
- Healthcare (Disease prediction, Medical image analysis)
- Finance (Fraud detection, Risk analysis)
- Education (Personalized learning)
- Agriculture (Crop prediction, Yield analysis)
- Content Creation (Text, Image, Video generation)

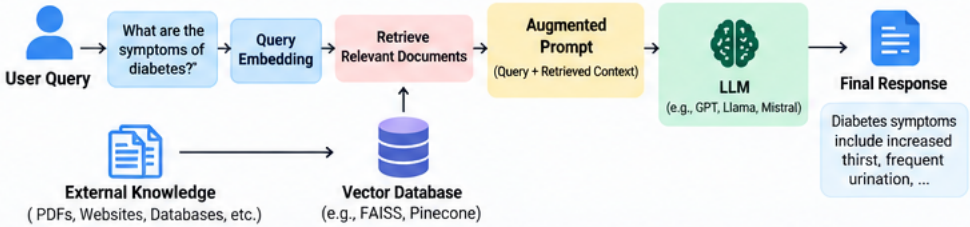
AI Workflow with Architecture

From Data to Real-World Application

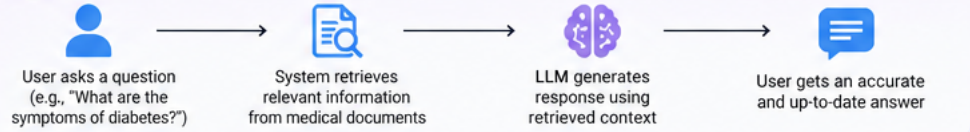
1 End-to-End AI Workflow with Architecture



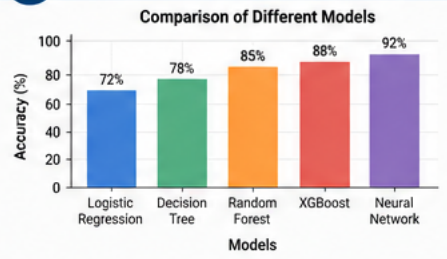
2 RAG (Retrieval Augmented Generation) Architecture



3 Example: Chatbot with RAG (Healthcare)



4 Model Performance (Example Graph)



5 Dataset Example (Sample Distribution)

